

# File Integrity Monitor

## Objective

To develop a Python application that monitors the integrity of files inside a selected folder by generating SHA-256 hashes and detecting whether files are added, modified, or deleted.

## Tools Used

- Python
- Pydroid 3
- hashlib
- os
- SHA-256 Hashing

## Working

The application scans the selected folder and creates a SHA-256 hash for every file. These hashes are stored as a baseline. After the user makes changes in the folder, the program scans it again and compares the new hashes with the previous ones. It reports newly added files, modified files, and deleted files, helping users detect unauthorized changes.

## Features

- File Integrity Monitoring
- SHA-256 Hash Generation
- Detects New Files
- Detects Modified Files
- Detects Deleted Files
- Fast Folder Scanning
- Simple Console Output

## Conclusion

This project demonstrates how cryptographic hashing can be used to verify file integrity. It is a simple cybersecurity tool that helps detect unexpected changes in important files and introduces the basic concept of File Integrity Monitoring (FIM), which is widely used in security and compliance.